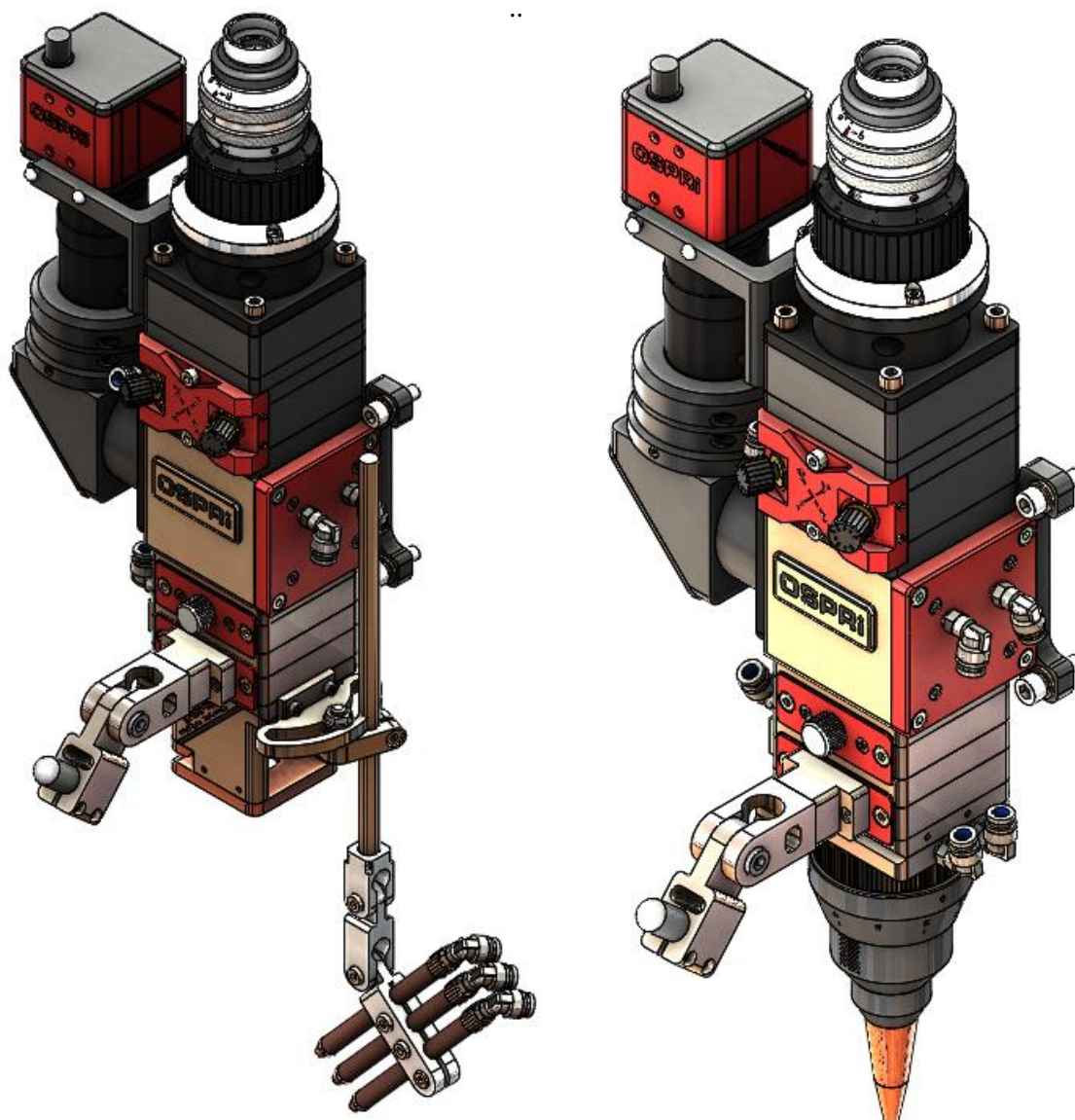


LW403 welding joint

User manual



Shenzhen Ospri Intelligent Technology Co., Ltd.

Tel.: 0755-85225225

Fax: 4008266163-19300

Mobile: 13915329830

Email: z.shi@sz-ospri.com

Address: Floor 2, Building C, No. 10 East Zone, Science and Technology Industrial City, Shangxue, Bantian, Longgang District, Shenzhen City

Foreword

Dear users:

Welcome to use LW403 welding joint produced by Shenzhen Ospri Intelligent Technology Co., Ltd. It is our great honor to gain your confidence in our products.

In order to make you have an overall view of the product, convenient for your use, we specifically provide the user manual for you, including product characteristics, structural feature, technical feature, direction for use, maintenance, etc. It's an essential guide when you use this product.

Please read the user manual carefully before use. I'm sure it will be helpful for you to use this product. In addition, if you have any questions during use, please contact us, and we will serve you wholeheartedly.

Special Statement:

The contents of the user manual are protected by copyright law, without approval of Shenzhen Ospri Intelligent Technology Co., Ltd, any organization or individual shall not copy or write by any means and forms.

To ensure your safety and regular work of the product, please read the instruction carefully before using our product.

Table of Contents

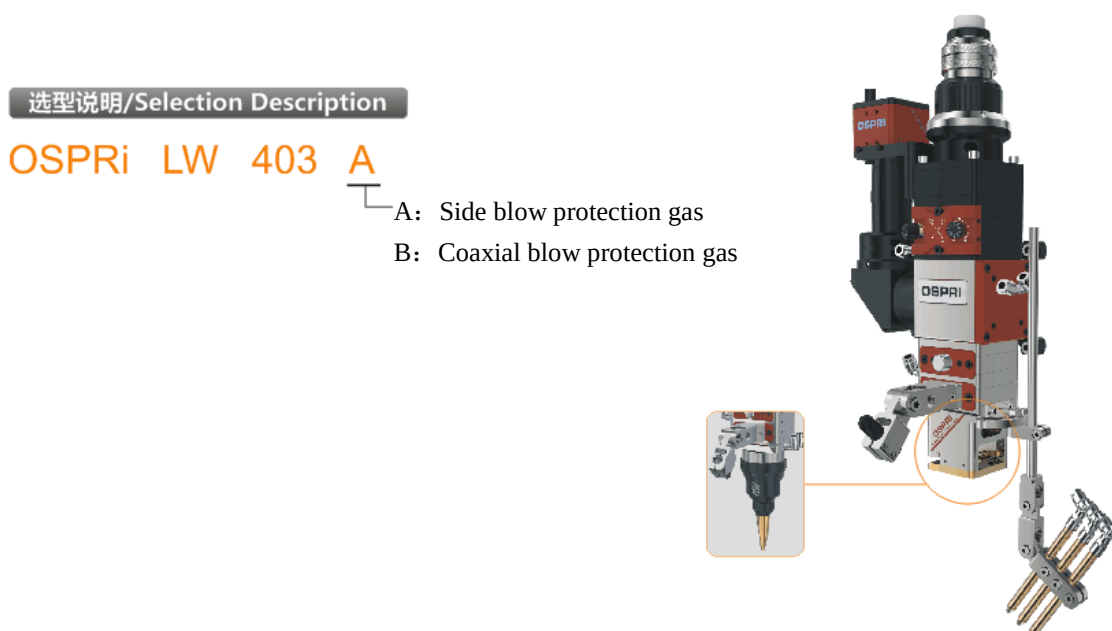
Table of Contents.....	3
Chapter 1 General.....	4
1.1 Principle of the product.....	4
1.2 Description of product selection.....	4
1.3 Product parameter	4
1.4 Matters needing attention.....	5
Chapter 2 Structural Characteristics	6
2.1 A brief description of the product mix	6
2.2 Brief description of product components.....	7
Chapter 3 Product Installation.....	8
3.1 installation of welding joint	8
3.2 Pipeline coupling	8
3.2.1 Cooling pipeline	8
3.2.2 Auxiliary gas pipeline	9
3.3 QBH installation.....	10
Chapter 4 Production debugging.....	13
4.1 Description of centering adjustment	13
4.1.1 Light beam centering.....	13
4.1.2 Collimating adjustment	14
Chapter 5 Maintenance.....	15
5.1 Maintenance of collimating lens	15
5.1.1 Disassembly of the lens of collimating lens.....	15
5.1.2 Cleaning of collimating lens	16
5.2 Maintenance and replacement of protective lens	17
5.2.1 Disassembly of protective lens.....	17
5.2.2 Cleaning of protective lens.....	18
5.3 Maintenance and replacement of focusing lens	18
5.4 Maintenance and replacement of reflector	19

Chapter 1 General

1.1 Principle of the product

Generated by the laser device and transmitted through the out light path, the laser is focused by focusing lens in the welding joint, then it acts on the weld joint between the work materials. With the aid of shielding gas (preventing material from being oxidized), the material is liquefied to form a specific molten bath, so as to achieve the purpose of welding.

1.2 Description of product selection



1.3 Product parameter

Parameters	Specifications
Laser Power	$\leq 3000\text{W}$
Focal length	180mm, 250 mm etc.
Paraxial nozzle diameter	$\leq 3\text{mm}$
Coaxial nozzle diameter	3Mm and 6mm
Focus adjustment range	None
Centering adjustment range	$\pm 2\text{mm}$

Auxiliary air pressure	<1.0MPa
Interface types of optical fiber	QBH
Applicable laser brands	Ruko and IPG, etc.

Table 1

1.4 Matters needing attention

① In order to ensure personal safety, when the welding joint is used with the laser welding machine, please wear special optical fiber laser protective goggles.

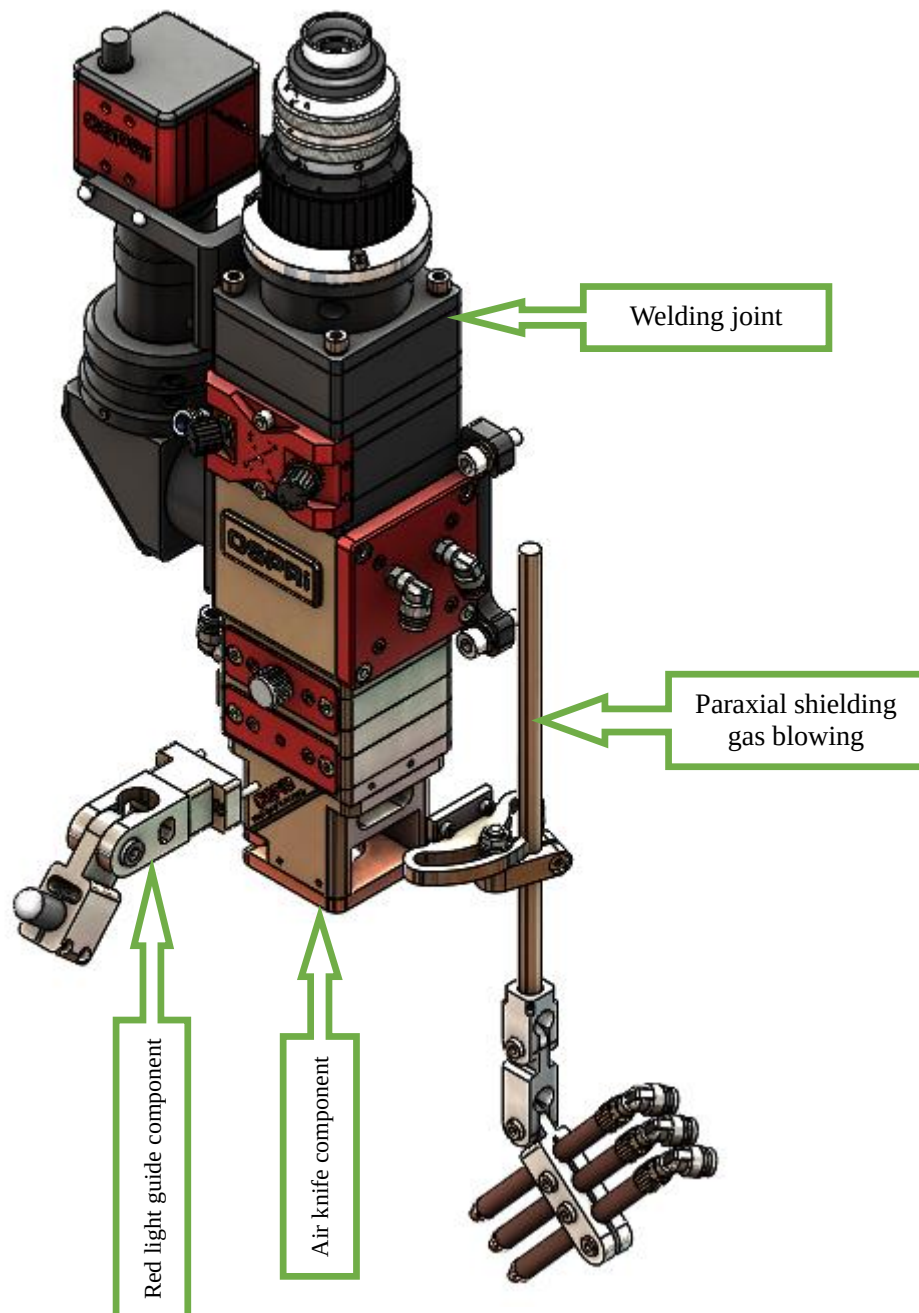
Caution: when laser is processing products, please use related protection device to avoid damage of the laser beam to the human body.



Chapter 2 Structural Characteristics

2.1 A brief description of the product mix

Because of dispatching and packing, the customer is required to assemble in accordance with this Schematic Diagram of Assembly 1 for LW403A (referring to LW403A for other types):



Warm remind : It is suggested that the air knife component is added when the power is greater than 1000W (the function of the air knife component is to prevent the spatter from falling on the protective lens during welding to protect the protective lens).

Fig.1

Abbreviated drawing of product mix containing the lens, as shown in Figure 2:

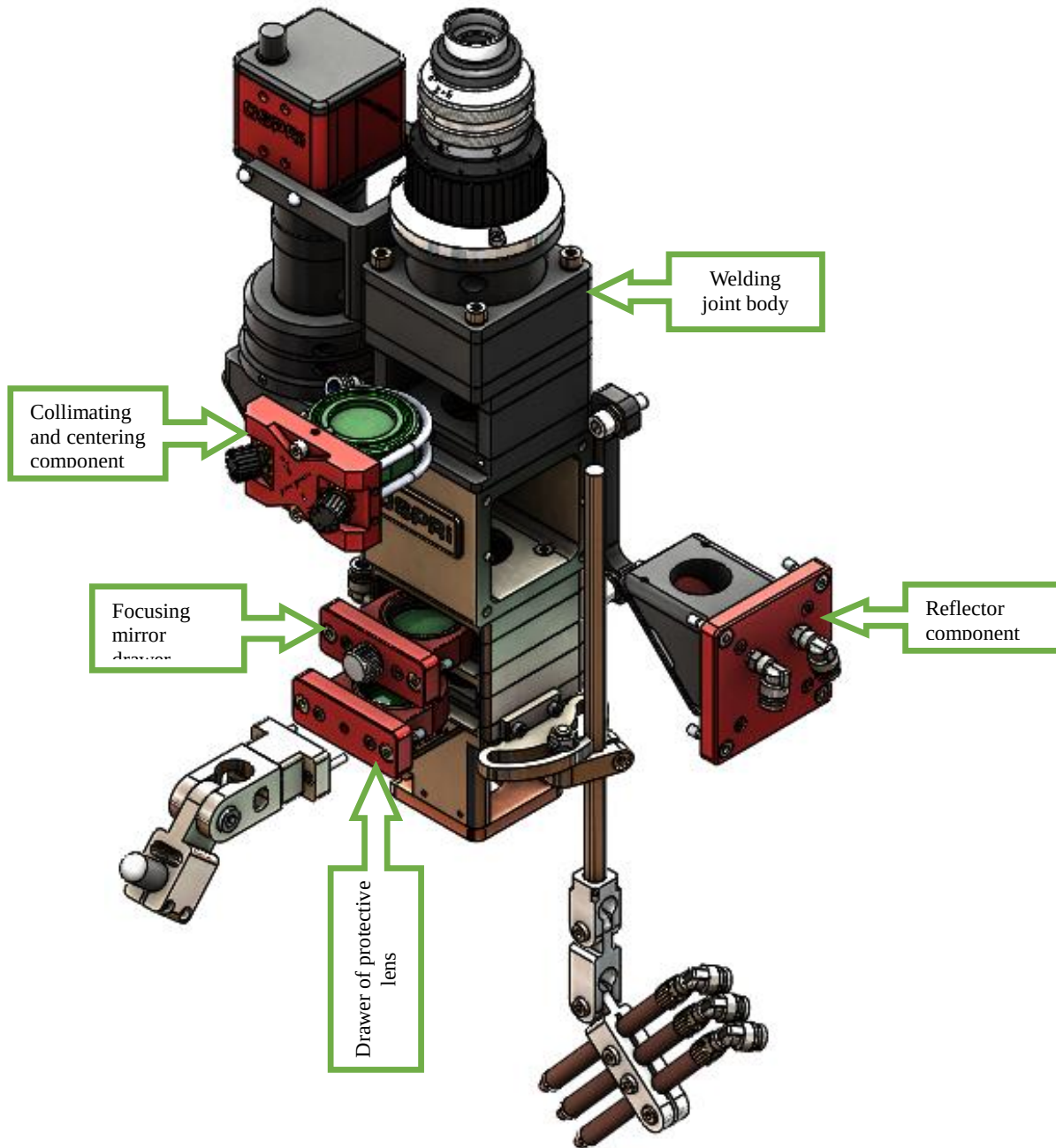


Figure 2

2.2 Brief description of product components

- ① Welding joint body: achieve QBH locking and fine tuning of beam-divergence angle;
- ② Collimating and centering component: achieve the overlap of the laser

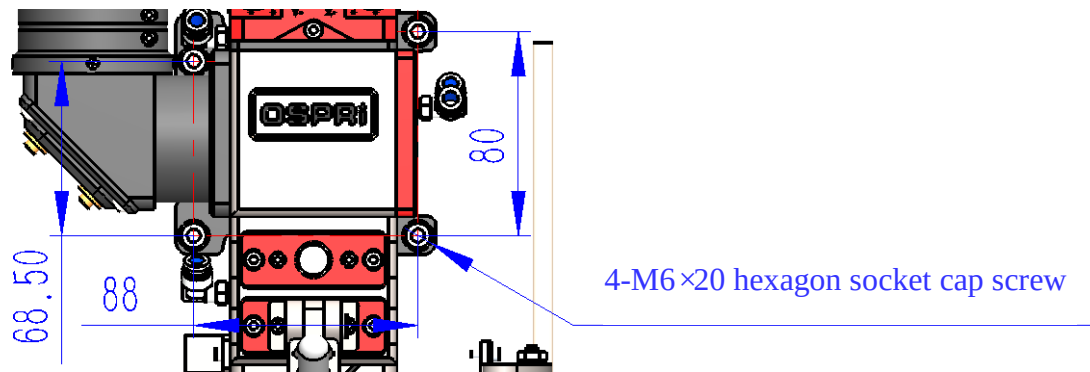
welding center and the crosshairs center of screen;

- ③ Drawer of focusing lens: replace the focusing lens with different focal lengths;
- ④ Drawer of protective lens: prevent the dust, etc from entering into a focusing lens for protection;
- ⑤ Reflector component: assist camera in imaging.

Chapter 3 Product Installation

3.1 installation of welding joint

- ① The welding joint and the equipment body are fixed with the mounting plate.

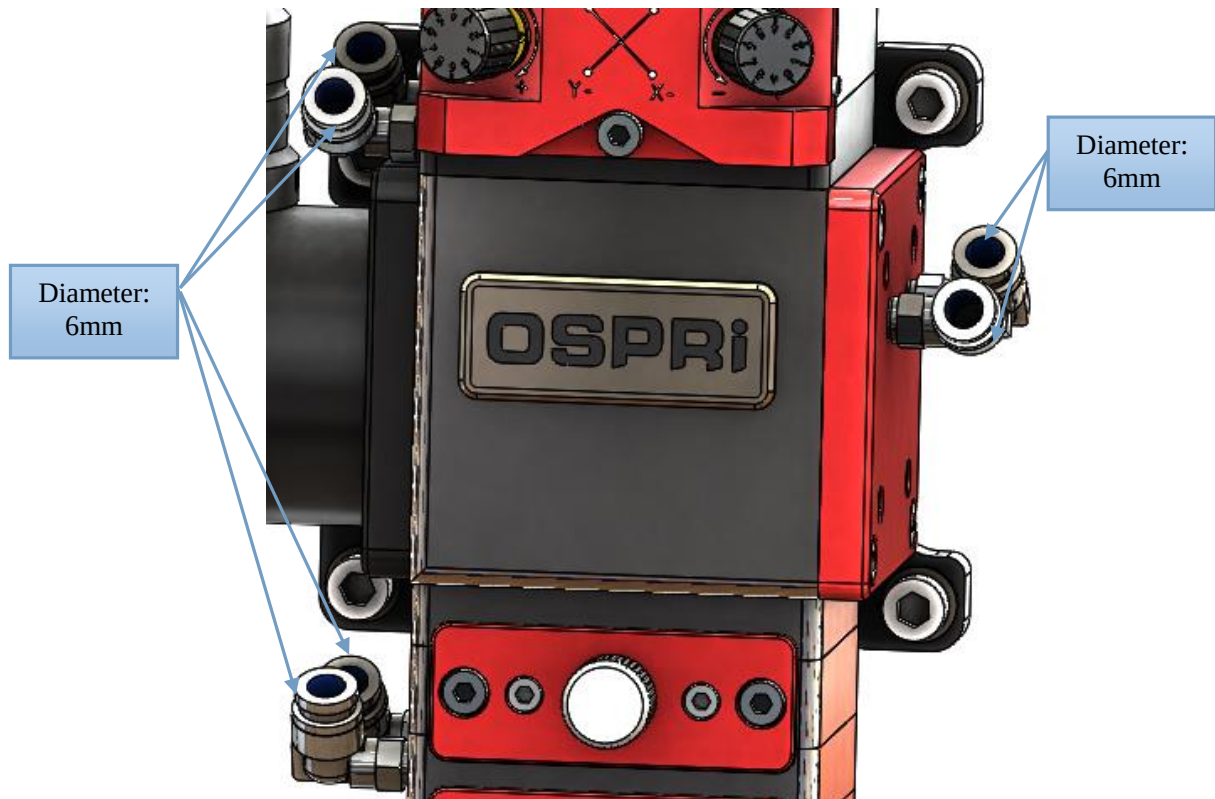


- ② There are 4* $\phi 6.5$ mounting holes on the mounting plate. The relative positions are shown in the figure.

3.2 Pipeline coupling

3.2.1 Cooling pipeline

The joints of collimating and centering component, reflector component, focus lens and drawer of protective lens are interconnected in turn.



3.2.2 Auxiliary gas pipeline

- ① Compressed air is connected to the 8mm air pipe (air knife, lateral blowing compressed air, to prevent the spatter from damaging the protective lens).
- ② The paraxial blowing protective gas is introduced into the 6mm air pipe, and the input pressure is $< 1.0\text{Mpa}$ (as shown in Figure 1).
- ③ The coaxial blowing protective gas is connected to the 6mm air pipe, and the input pressure is $< 1.0\text{Mpa}$ (as shown in Figure 2).



Fig.1

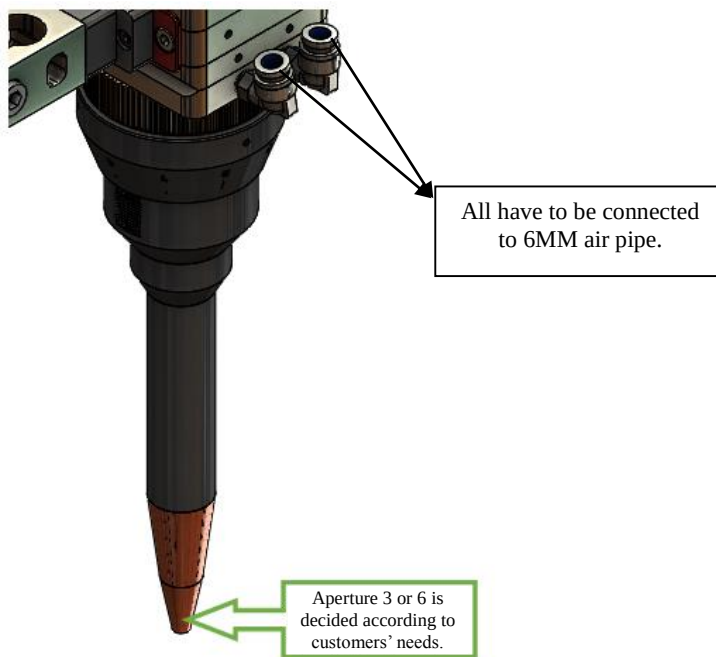
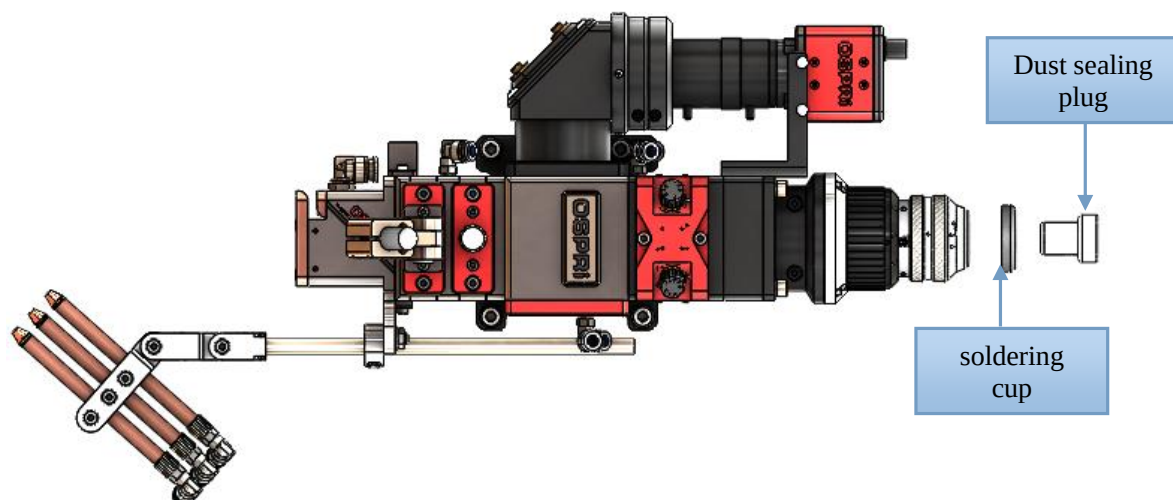


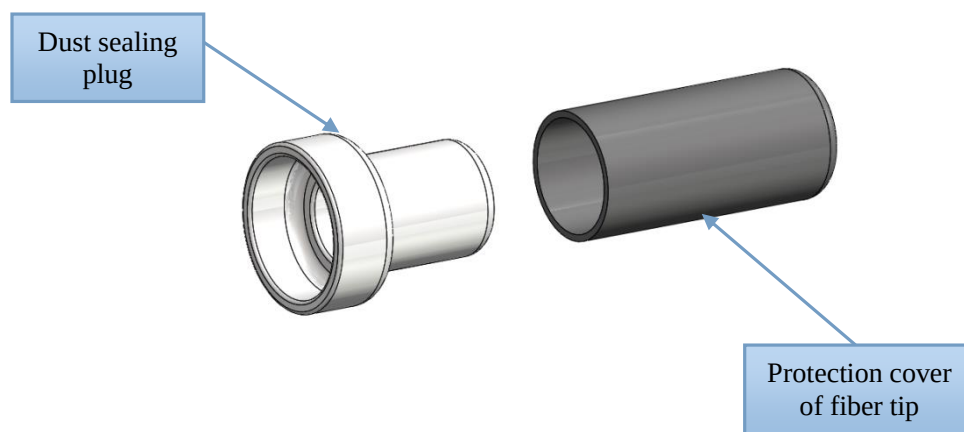
Figure 2

3.3 QBH installation

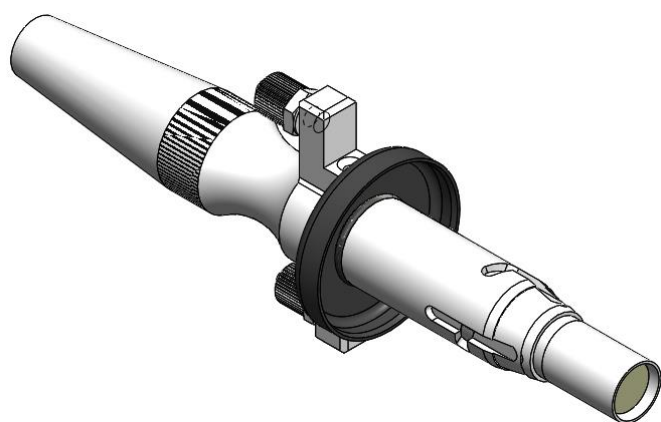
① Place the fiber optic welding joint horizontally and remove the dust sealing plug and dust-proof cover.



② Insert the dust sealing plug removed into the protection cover of fiber tip.



③ Place the dust cover removed into the fiber tip. As shown in the following figure:

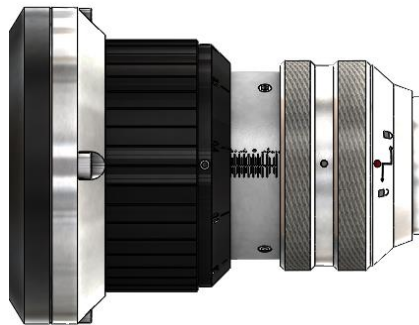




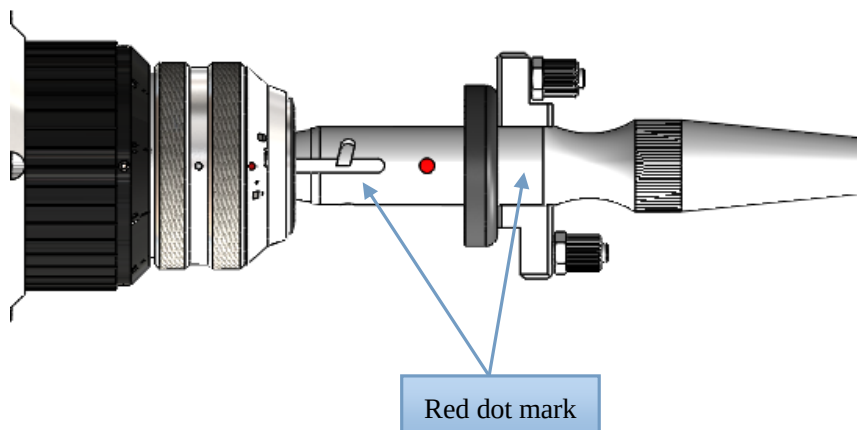
Attention: If a dust-proof pad is originally equipped on the laser head, the original dust-proof pad should still be installed when it is mounted.

Otherwise, the airtightness may be affected. It also causes that parallel light turns into divergent light, affecting the welding effect.

④Screw the QBH connector to the open state: counterclockwise revolve it to the limiting position (you can hear "thump"). (Counterclockwise rotation to the direction of the "unlock" icon)



⑤ Aim the red dot on the fiber tip at the red dot of the QBH connector and slowly insert the fiber tip into the QBH connector.



⑥ Screw the QBH connector to the locking state: revolve it to the limiting position clockwise (you can hear "thump"), and lift upward and revolve the nut, and then revolve clockwise the nut again until it compresses the fiber tip.

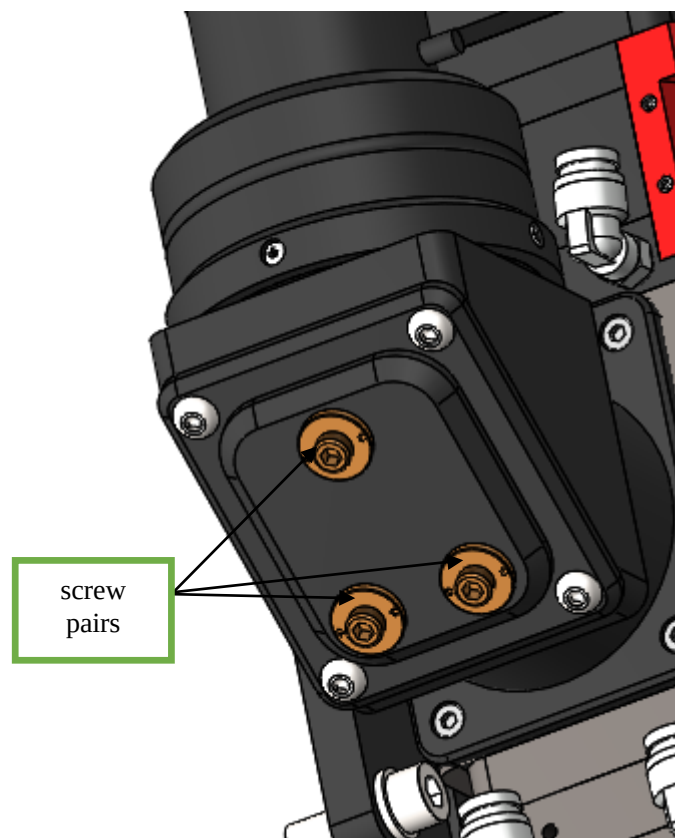
(Clockwise rotation to the direction of the "lock" icon)

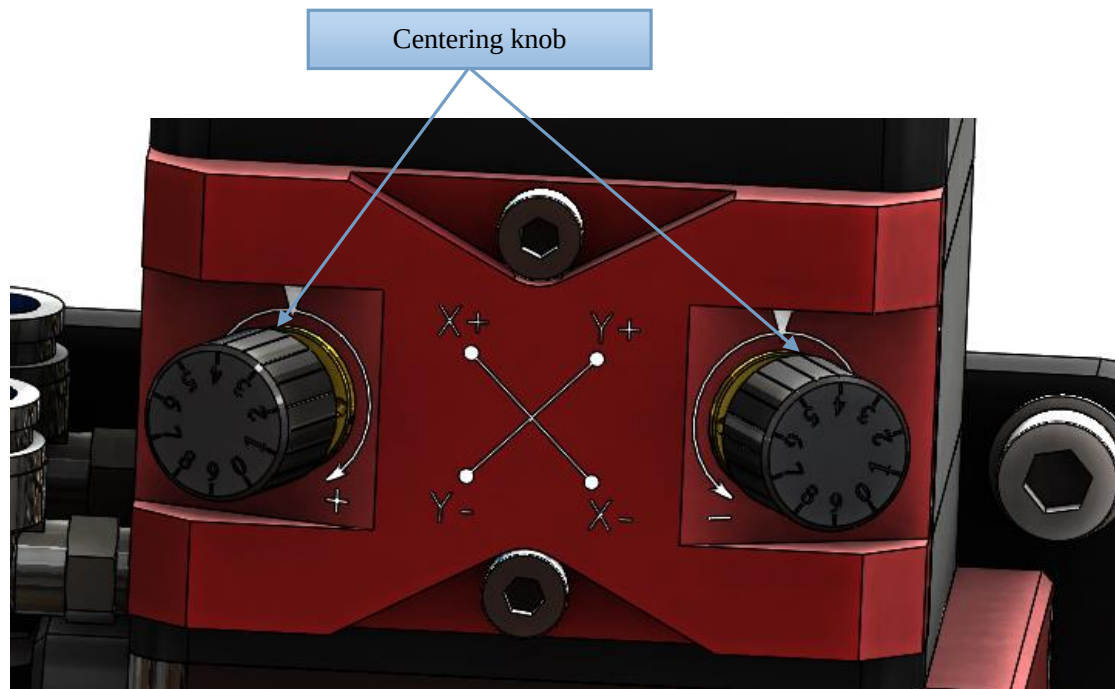
Chapter IV Production debugging

4.1 Description of centering adjustment

4.1.1 Light beam centering

In order to achieve good welding and teaching effect, the laser beam must overlap with the cross line of the screen. When it deviates from the center of the cross line of the screen, it needs to be centered by CCD component. If the laser center is not what you want, make the center of laser beam overlap the cross line of screen through regulating the centering component and CCD component (Generally, it is regulated properly before leaving the factory. If the laser beam of the welding joint you receive doesn't overlap the center of the cross line of screen, please make the laser center overlap the center of the cross line of screen by adjusting the three screw-thread secondary adjusting lenses on the CCD component).





The relation between the beam position and the adjusting knob

- ① There are 2 centering knobs.
- ② Adjustment manner: adjust two knobs, back and forth, to achieve the purpose of centering.

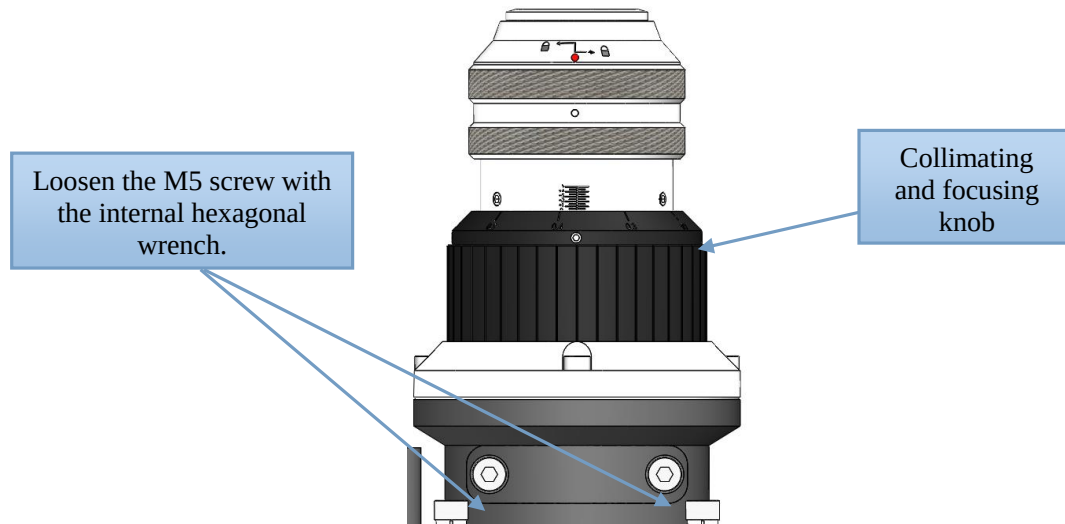
4.1.2 Collimating adjustment

In order to make the laser beam through the focus lens become parallel light, it is necessary to adjust the collimating scale to the theoretical collimating focal length.

Scale of the theoretical collimating focal length of the optical fiber welding joint LW403A is 0 . (Generally, it is adjusted properly and locked before leaving the factory.)

Adjustment method

- ① Loosen two M5 hexagon socket cap screws on the collimating and locking component, and revolve the focusing knob to the desired position, so that the light spot is of the theoretical value, that is, the energy is the most concentrated.



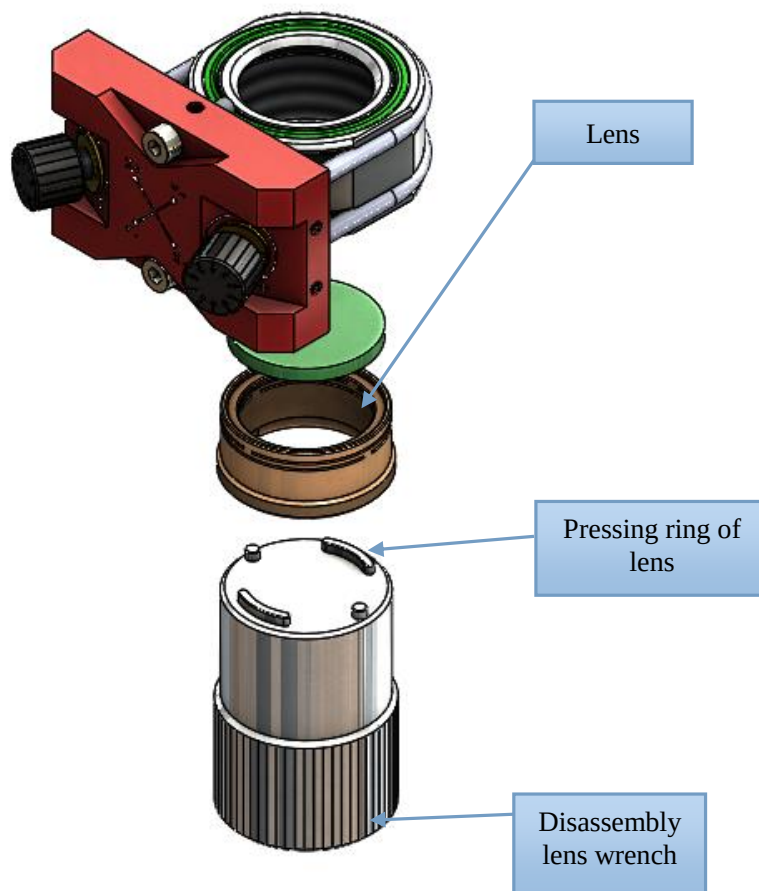
Chapter V Maintenance

5.1 Maintenance of collimating lens

The collimating lens is located in the collimating and centering component, and it can be maintained periodically, and the recommended cleaning cycle is 1 to 2 months. When cleaning and replacing, make sure that the operating space is clean and dust free, so as not to pollute the lenses.

5.1.1 Disassembly of the lens of collimating lens

① First, remove 2 M4*16 fastening screws with an internal hexagonal wrench and take out the collimating and centering component.



- ② Insert the pressing ring of lens with the wrench dismounting the lens, screw off the locking ring counterclockwise with the hand tool dismounting the lens.
- ③ Remove the pressing ring of lens and collimating lens in turn.

5.1.2 Cleaning of collimating lens

- ① Serviceable tools: dust-free wiping cotton swab, isopropyl alcohol and rubber air-blowing.
- ② Sprinkle the isopropyl alcohol onto the dust-free wiping cotton swab.
- ③ Gently pinch the side edge of the collimating lens with a thumb and forefinger of the left hand.
- ④ Hold the dust-free wiping cotton swab with right hand to gently wipe both sides of lens in a single direction from the bottom to the top or from left to right, and blow the lens surface with the rubber air-blowing, and confirm that there is not any foreign body on the cleansed lens surface.

⑤ The cleansed collimating lens must be installed into the collimating and centering component as soon as possible and inserted into the welding joint body or stored in other clean airtight containers.



Installation direction of collimating lens:

Collimating lens: flat surface is directed at the laser head and convex surface is directed at the focusing lens.

Focusing lens: flat surface is directed at the protective lens and convex surface is directed at the collimating lens.

5.2 Maintenance and replacement of protective lens

The protective lens is located below the focusing drawer component. When the protective lens has impurities or foreign body attached, the protective lens will be damaged because impurities or foreign body absorb the laser-heating, therefore, you need to clean the protective lenses regularly. Recommended cleaning cycle: a week. Meanwhile, protective lens is a vulnerable part, and the damaged lens needs to be replaced.

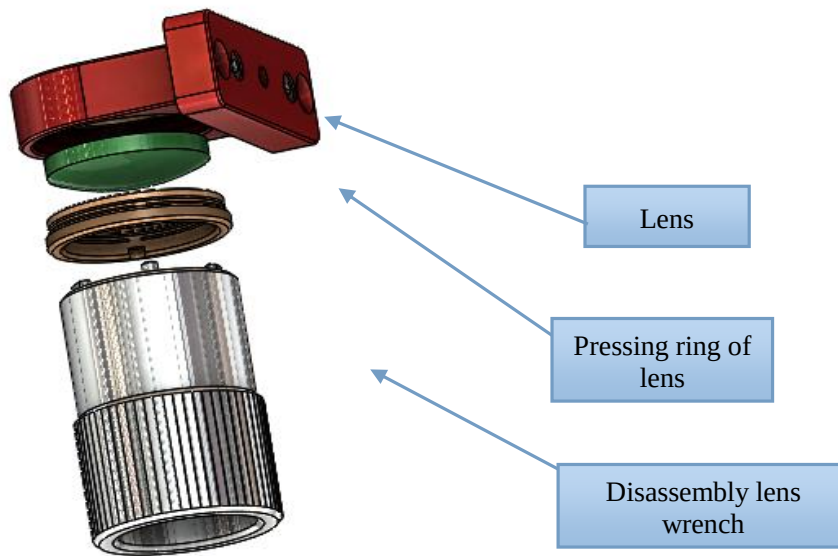


Attention:

When cleaning and changing the protective lens, avoid the oil stain on the hand or the dust in the environment to pollute the productive lens.

5.2.1 Disassembly of protective lens

① Use a wrench to remove the red light guide component, slowly and smoothly pull out the protective lens drawer component and move it into a clean and dust-free environment.



② Insert the pressing ring of lens with a wrench dismounting the lens, and screw off the locking ring counterclockwise with a handheld wrench dismounting the lens.

③ Remove the pressing ring of lens and protective lens in turn.

5.2.2 Cleaning of protective lens

① Serviceable tools: dust-free wiping cotton swab, isopropyl alcohol and rubber air-blowing.

② Sprinkle the isopropyl alcohol onto the dust-free wiping cotton swab.

③ Gently pinch the side edge of the protective lens with a thumb and index finger of the left hand.

④ Hold the dust-free wiping cotton swab with right hand to gently wipe both sides of lens in a single direction from the bottom to the top or from left to right, and blow the lens surface with the rubber air-blowing, and confirm that there is not any foreign body on the cleansed lens surface.

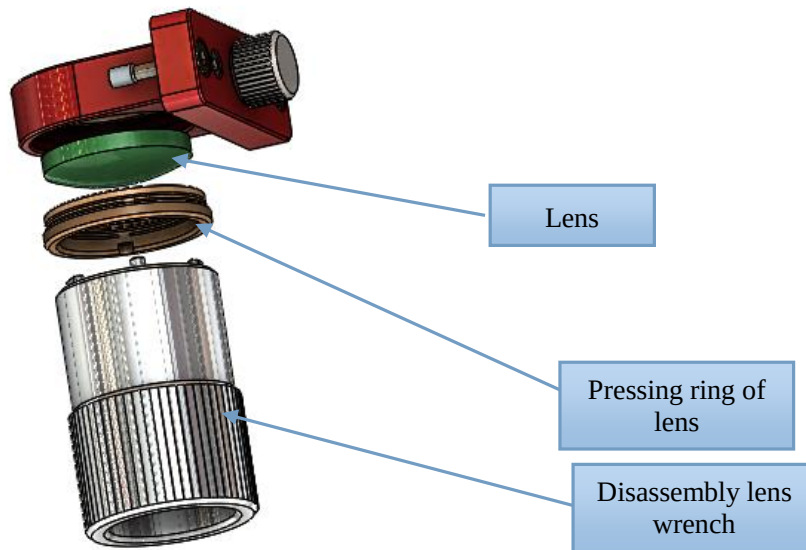
⑤ The cleansed productive lens must be installed into the productive drawer as soon as possible and inserted into the welding joint body or stored in other clean airtight containers.

5.3 Maintenance and replacement of focusing lens

The focusing lens drawer is located on the protective lens drawer and among

the reflector component. The focusing lens is located inside the focusing lens drawer component. It can be maintained regularly, and the recommended cleaning cycle is 1 to 2 months. When cleaning and replacing, make sure that the operating space is clean and dust free, so as not to pollute the lenses.

1. Remove 2-M4x16 screws with a wrench, and pull out the focusing lens drawer slowly and smoothly, and move it into a clean and dust-free environment.



2. Insert the pressing ring of lens with a wrench dismounting the lens, and screw off the locking ring counterclockwise with a handheld wrench dismounting the lens.
3. Remove the pressing ring of lens and focusing lens in turn.

5.4 Maintenance and replacement of reflector

The reflector component is located between the collimating and centering component and the focusing lens drawer component. The reflector lens is located inside the reflector component. It can be maintained periodically, and the recommended cleaning cycle is 1 to 2 months. When cleaning and replacing, make sure that the operating space is clean and dust free, so as not to pollute the lenses.

1. Remove 4-M4x16 with a internal hexagonal wrench, and pull out the

reflector component from the welding body and move it in a clean and dust-free environment.

2. Remove 4-M3x8 countersunk socket cap screws with internal hexagonal wrench.
3. Remove the gland and lens of the reflector in turn.
4. When installing the reflector, it is important to note the installation direction (arrow points outward).

